Saptak Sil

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Education

Indian Institute of Technology (IIT), Kharagpur **June 2025 – June 2030**
Ph.D. in Computer Science

Jadavpur University **September 2023 – June 2025**
M.E. in Information Technology (IT) **CGPA: 8.28**

Institute of Engineering and Management **August 2018 – June 2022**
B.Tech in Electronics and Communication (ECE) **CGPA: 9.43**

Hariyana Vidya Mandir, Kolkata **Graduated: 2018**
12^{th} Standard (PCMB, CBSE) **81.6%**

Calcutta Boys' School **Graduated: 2016**
10^{th} Standard (All Subjects, ICSE) **90.2%**

Experience

IIT Kharagpur June 2025 – Present
Junior Researcher – Working on 3D Computer Vision

Cognizant Technology Solutions Aug 2022 – Sept 2023
Junior Software Developer — Java, HTML, CSS, JS, SQL *Kolkata, India*

- Involved in the **analysis**, **design**, **implementation**, and support of Insurance Software Engineering products.
- Conducted **code reviews** to ensure quality and adherence to standards.
- Thoroughly **tested** applications after production release to identify and resolve issues.
- Reduced the **ticket size by 15%**.

Technical Skills

Programming Languages: Python,Java, C
Libraries and Frameworks::Pandas, NumPy, Pytorch, Keras, Scikit-learn, Matplotlib, Seaborn
Databases: MySQL
Tools: Git, GitHub, Google Colab
Concepts:Data Analysis, Machine Learning, Data Visualization

Achievements

- Solved **300+ Data Structures and Algorithmic** problems on coding platform such as **LeetCode** and GFG.
- Contest Rating: **1470 @**LeetCode
- Gate Score CS(2023) **- 307**
- Gate Score DA(2025) **- 319**

Area of interest

- Machine Learning & Artificial Intelligence
- Programming, Algorithms and Data Structures
- Digital Logic and Design

Publications

Detecting Profile Cloning on Online Social Networks with Machine Learning (Accepted in CICBA)

- Focused on identifying and mitigating threats related to privacy and security in online social networking platforms. Developed and implemented machine learning algorithms to enhance data protection and user privacy, tackling emerging security challenges in a digital ecosystem.
- Achieved an accuracy of 95.8% for a 4-class classification task and 90.5% for a 2-class classification task, effectively addressing emerging security challenges.

Fake Profile Detection on Online Social Networks Using Machine Learning (Published in JoCAAA)

* We focus on detecting malicious behaviors on Twitter through the application of various machine learning algorithms. By conducting extensive experiments and performance evaluations, we compare traditional machine learning techniques with deep learning models. Our results demonstrate that deep learning approaches, particularly those leveraging neural network architectures, significantly outperform classical methods, achieving higher accuracy in identifying fake profiles and offering more robust solutions for OSN security.

Projects

Customer Churn Prediction Using ANN **[Colab Notebook]**

- Preprocessed customer data, including feature selection and one-hot encoding of categorical variables.
- Developed an Artificial Neural Network (ANN) using TensorFlow/Keras, incorporating multiple hidden layers and activation functions to capture complex customer behavior patterns.
- Achieved a training accuracy of 83.01% and a testing accuracy of 78.82%, demonstrating strong predictive performance.

Diabetes Prediction Using Machine Learning

- Preprocessed patient data, applied normalization, and split the dataset into 70% training and 30% testing subsets.
- Implemented and evaluated multiple machine learning algorithms, with Linear Regression achieving the highest accuracy among the models tested.